

Nmap: Ultimate Command Cheat-sheet

Basic form: `nmap [scan type(s)] [options] {target}`

Example target names I'll use: `scanme.nmap.org` , `192.168.1.1` , `192.168.1.0/24` , `10.0.0.5` .

TARGET SPECIFICATION

These control *what* Nmap scans.

- Hostnames, IPs, networks
 - Example: `nmap scanme.nmap.org` — scans that hostname.
 - Example: `nmap 192.168.1.1` — scan single host.
 - Example: `nmap 192.168.1.0/24` — scan whole /24 network.
 - `iL <inputfilename>` — read targets from a file (one per line).
 - Example: `nmap -iL targets.txt`
 - Use when you have many targets to scan repeatedly.
 - `iR <num hosts>` — scan that many randomly chosen IPs.
 - Example: `nmap -iR 100` — scans 100 random IPv4 addresses.
 - `-exclude <host1[,host2],...>` — exclude specific hosts/networks.
 - Example: `nmap 192.168.1.0/24 --exclude 192.168.1.5,192.168.1.10`
 - Useful to omit known safe systems.
 - `-excludefile <exclude_file>` — exclude hosts listed in file.
 - Example: `nmap -iL allhosts.txt --excludefile skipme.txt`
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HOST DISCOVERY

Control how Nmap determines whether hosts are up.

- **sL** — List Scan (no probes). Just resolve names and show targets.
 - Example: `nmap -sL 192.168.1.0/28`
 - Use to verify which hosts WOULD be scanned (no network traffic).
- **sn** — Ping Scan (host discovery only, no port scan).
 - Example: `nmap -sn 192.168.1.0/24`
 - Shows which hosts are up without scanning ports.
- **Pn** — Treat all hosts as online (skip discovery).
 - Example: `nmap -Pn 10.0.0.5`
 - Useful when ICMP is blocked or you want to force port scanning.
- **PS/PA/PU/PY[portlist]** — TCP SYN / TCP ACK / UDP / SCTP discovery to given ports.
 - Example: `nmap -PS22,80 192.168.1.0/24`
 - Sends SYN probes to listed ports to discover hosts.
- **PE/PP/PM** — ICMP echo/timestamp/netmask request for discovery.
 - Example: `nmap -PE 10.0.0.0/24`
 - Traditional ICMP pings to detect live hosts.
- **PO[protocol list]** — IP protocol ping (ICMP, GRE, etc).
 - Example: `nmap -PO 192.168.1.1`
 - Probes specific IP protocols to detect hosts.
- **n / R** — DNS resolution controls.
 - **n**: never do DNS resolution (speeds scan).
 - Example: `nmap -n 192.168.1.0/24`
 - **R**: always resolve names.
 - Example: `nmap -R scanme.nmap.org`
- **-dns-servers <serv1[,serv2]...>** — use specific DNS servers.

- Example: `nmap --dns-servers 8.8.8.8 scanme.nmap.org`
 - `-system-dns` — use OS resolver for DNS lookups (may follow OS config).
 - Example: `nmap --system-dns scanme.nmap.org`
 - `-traceroute` — run traceroute to each host after scan.
 - Example: `nmap --traceroute scanme.nmap.org`
 - Shows network path hops to the target.
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SCAN TECHNIQUES

Which port/probe method to use.

- `sS` — TCP SYN scan (stealthy, sends SYN then resets).
 - Example: `nmap -sS 192.168.1.1`
 - Default and fast; requires raw sockets (privileged).
- `sT` — TCP Connect() scan (uses OS connect()).
 - Example: `nmap -sT 192.168.1.1`
 - Use when unprivileged or raw sockets not available.
- `sA` — TCP ACK scan (useful to map firewall rules).
 - Example: `nmap -sA 192.168.1.1`
 - Helps determine if firewall is stateful / filters.
- `sW` — TCP Window scan (variation of ACK scan).
 - Example: `nmap -sW 192.168.1.1`
- `sM` — Maimon scan (rare; exploits TCP behavior).
 - Example: `nmap -sM 192.168.1.1`
- `sU` — UDP scan.
 - Example: `nmap -sU -p 53,123 192.168.1.1`
 - UDP scanning is slow and may require higher timeouts (-T, --max-retries).

- `sN/-sF/-sX` — TCP Null / FIN / Xmas scans (stealthy to some stacks).
 - Example: `nmap -sX 192.168.1.1`
 - Send unusual flag combinations; used for evasion on older systems.
 - `-scanflags <flags>` — custom TCP flags.
 - Example: `nmap --scanflags SYN,ACK -p 80 192.168.1.1`
 - `sl <zombie host[:probeport]>` — Idle scan (spoofed scanning via zombie).
 - Example: `nmap -sl zombie.example.com 192.168.1.1`
 - Asynchronous, anonymous scanning (complex; requires suitable zombie).
 - `sY` / `sZ` — SCTP INIT / COOKIE-ECHO scans for SCTP ports.
 - Example: `nmap -sY 192.168.1.1`
 - `sO` — IP protocol scan (determine which IP protocols supported).
 - Example: `nmap -sO 192.168.1.1`
 - `b <FTP relay host>` — FTP bounce scan (historical, rarely works).
 - Example: `nmap -b ftp.example.com 192.168.1.1`
 - Relies on misconfigured FTP servers; uncommon now.
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PORT SPECIFICATION AND SCAN ORDER

Control which ports Nmap checks.

- `p <port ranges>` — specify ports to scan.
 - Example: `nmap -p22,80,443 192.168.1.1`
 - Or full: `p1-65535` to scan all TCP ports.
- Example with mixed protocols: `p U:53,111,T:21-25,80,139,8080,S:9`
 - `U:` = UDP ports, `T:` = TCP ports, `S:` = SCTP ports.
- `-exclude-ports <port ranges>` — skip certain ports.
 - Example: `nmap -p1-1024 --exclude-ports 22,80 192.168.1.1`

- **F** — Fast mode (scan fewer commonly-used ports).
 - Example: `nmap -F 192.168.1.1`
 - Good for quick reconnaissance.
 - **r** — scan ports sequentially (no randomization).
 - Example: `nmap -r -p1-100 192.168.1.1`
 - Helpful if you expect IDS to detect random port order.
 - **-top-ports <number>** — scan the top N common ports.
 - Example: `nmap --top-ports 100 192.168.1.0/24`
 - **-port-ratio <ratio>** — scan ports more common than ratio.
 - Example: `nmap --port-ratio 0.5 192.168.1.1` — scans ports with popularity > 50%.
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SERVICE / VERSION DETECTION

Probe open ports to determine service and version strings.

- **sV** — probe services to detect versions.
 - Example: `nmap -sV -p22,80 192.168.1.1`
 - Tries application-layer probes to identify software/version.
 - **-version-intensity <level>** — 0 (light) to 9 (aggressive).
 - Example: `nmap --version-intensity 0 -sV 192.168.1.1` (fast & light)
 - **-version-light** — shortcut for low intensity (≈ 2).
 - Example: `nmap --version-light -sV scanme.nmap.org`
 - **-version-all** — try all probes (intensity 9).
 - Example: `nmap --version-all -sV 192.168.1.1`
 - More accurate, slower and noisier.
 - **-version-trace** — show details of version probe exchanges (debugging).
 - Example: `nmap --version-trace -sV 192.168.1.1`
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SCRIPT SCAN (NSE — Nmap Scripting Engine)

Run Lua scripts for discovery, vulnerability checking, brute force, etc.

- `-sC` — run default script category (equivalent to `-script=default`).
 - Example: `nmap -sC scanme.nmap.org`
 - Good quick check of common findings.
- `-script=<Lua scripts>` — provide script names, files or categories.
 - Example: `nmap --script vuln,auth 192.168.1.1`
 - Many categories: `discovery`, `vuln`, `auth`, `malware`, etc.
- `-script-args=<n1=v1,[n2=v2,...]>` — arguments to scripts.
 - Example: `nmap --script smb-brute --script-args userdb=users.txt,passdb=passes.txt 192.168.1.1`
- `-script-args-file=filename` — script args from file.
 - Example: `nmap --script smb-brute --script-args-file args.txt 192.168.1.1`
- `-script-trace` — show script traffic (debugging).
 - Example: `nmap --script-trace --script http-enum scanme.nmap.org`
- `-script-updatedb` — update local script database (after adding scripts).
 - Example: `nmap --script-updatedb`
- `-script-help=<Lua scripts>` — show help for scripts.
 - Example: `nmap --script-help http-vuln-cve2006-3392`

Note: specify script files or categories separated by commas.

OS DETECTION

- `-O` — enable OS detection.
 - Example: `nmap -O scanme.nmap.org`
 - Nmap sends probes and uses fingerprint database to guess OS.

- `-osscan-limit` — only OS-detect promising targets (speeds up large scans).
 - Example: `nmap --osscan-limit 192.168.1.0/24`
- `-osscan-guess` — be more aggressive in guessing when match is uncertain.
 - Example: `nmap --osscan-guess scanme.nmap.org`

OS detection can be noisy and is less reliable on filtered hosts.

TIMING AND PERFORMANCE

Tweak speed vs accuracy / stealth.

- `T<0-5>` — timing template (0 slowest, 5 fastest).
 - Example: `nmap -T4 192.168.1.0/24` — faster, typical for local scans.
 - Use `T1` or `T0` when stealth needed or networks are slow.
- `-min-hostgroup / --max-hostgroup <size>` — host group size for parallel scanning.
 - Example: `nmap --min-hostgroup 16 --max-hostgroup 64 192.168.1.0/24`
- `-min-parallelism / --max-parallelism <numprobes>` — probe parallelization.
 - Example: `nmap --max-parallelism 100 192.168.1.0/24`
- `-min-rtt-timeout / --max-rtt-timeout / --initial-rtt-timeout <time>` — adjust RTT timeouts.
 - Example: `nmap --initial-rtt-timeout 200ms 10.0.0.0/24`
- `-max-retries <tries>` — cap number of retransmissions for probes.
 - Example: `nmap --max-retries 2 scanme.nmap.org`
- `-host-timeout <time>` — abandon a host after this time.
 - Example: `nmap --host-timeout 5m 192.168.1.0/24`
- `-scan-delay / --max-scan-delay <time>` — delay between probes.
 - Example: `nmap --scan-delay 50ms 192.168.1.1`
- `-min-rate <number> / --max-rate <number>` — global probes per second limits.
 - Example: `nmap --min-rate 100 192.168.1.0/24` — at least 100 packets/sec.

Adjust these when scanning large networks or when trying to avoid overwhelming the target.

FIREWALL / IDS EVASION AND SPOOFING

Options to change packet shapes, source IPs, fragments, etc. **Use ethically and only with permission.**

- `f / -mtu <val>` — fragment packets or set MTU.
 - Example: `nmap -f 192.168.1.1`
 - Fragmentation can evade some packet filters (less effective today).
- `D <decoy1,decoy2[,ME],...>` — cloak your scan with decoys.
 - Example: `nmap -D RND:10 192.168.1.1` — use 10 random decoys plus you.
 - Makes attribution harder; noisy and may violate laws/policies.
- `S <IP_Address>` — spoof source IP (requires network-level control).
 - Example: `nmap -S 1.2.3.4 192.168.1.1`
 - Often not practical; replies won't return to you without MITM.
- `e <iface>` — use specified interface.
 - Example: `nmap -e eth0 192.168.1.1`
- `g/--source-port <portnum>` — use specific source port (may bypass rudimentary filters).
 - Example: `nmap -g 53 192.168.1.1` — source port 53 (DNS) sometimes allowed.
- `-proxies <url1,[url2],...>` — relay through HTTP/SOCKS4 proxies.
 - Example: `nmap --proxies socks4://127.0.0.1:9050 scanme.nmap.org`
 - Useful for TCP connect scans via proxy; not for raw packet scans.
- `-data <hex string> / -data-string <string> / -data-length <num>` — append custom payload.
 - Example: `nmap --data-string "HELLO" -p 80 192.168.1.1`
- `-ip-options <options>` — set IP options fields.

- Example: `nmap --ip-options "rr" 192.168.1.1` — record route IP option.
- `-ttl <val>` — set IP TTL.
 - Example: `nmap --ttl 64 192.168.1.1` — set packet TTL to 64.
- `-spoof-mac <mac/prefix/vendor>` — spoof MAC address.
 - Example: `nmap --spoof-mac 0A:0B:0C:0D:0E:0F 192.168.1.1`
 - Or `-spoof-mac Cisco` to select a vendor OUI.
- `-badsum` — send packets with bogus checksums (to confuse or test stacks).
 - Example: `nmap --badsum 192.168.1.1`
 - Many stacks drop such packets, so results vary.

Legal/ethical note: many of these options are noisy and could violate law/policy. Only use with explicit authorization.

OUTPUT OPTIONS

How Nmap prints and saves output.

- `-oN <file>` — normal human-readable output file.
 - Example: `nmap -oN results.txt scanme.nmap.org`
- `-oX <file>` — XML output.
 - Example: `nmap -oX results.xml scanme.nmap.org`
- `-oS <file>` — “s|<rlpt klldi3” (historic; rarely used) — typically ignore.
- `-oG <file>` — Greppable format (easier for parsing).
 - Example: `nmap -oG greppable.txt 192.168.1.0/24`
- `-oA <basename>` — write all three major formats at once (`basename.nmap` , `.xml` , `.gnmap`).
 - Example: `nmap -oA myscan 192.168.1.0/24`
- `-v` / `-vv` — increase verbosity.
 - Example: `nmap -v scanme.nmap.org`

- `d` / `dd` — increase debugging output.
 - Example: `nmap -d scanme.nmap.org` — very chatty; for troubleshooting.
 - `-reason` — show why a port is classified open/filtered/closed.
 - Example: `nmap --reason -p80 scanme.nmap.org`
 - `-open` — only show open (or possibly open) ports in output.
 - Example: `nmap --open scanme.nmap.org`
 - `-packet-trace` — dump all packets sent/received (very verbose).
 - Example: `nmap --packet-trace -sS scanme.nmap.org`
 - `-iflist` — list host network interfaces and routing (debugging).
 - Example: `nmap --iflist`
 - `-append-output` — append to existing output files (don't overwrite).
 - Example: `nmap --append-output -oN scan.log scanme.nmap.org`
 - `-resume <filename>` — resume an aborted scan using saved XML data.
 - Example: `nmap --resume myscan.xml`
 - `-noninteractive` — disable runtime keyboard interactions (useful for scripts).
 - Example: `nmap --noninteractive -oA myscan 192.168.1.0/24`
 - `-stylesheet <path/URL>` / `-webxml` / `-no-stylesheet` — control XSL for XML→HTML transforms.
 - Example: `nmap --webxml -oX myscan.xml scanme.nmap.org`
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MISCELLANEOUS

- `6` — enable IPv6 scanning.
 - Example: `nmap -6 -sS -p 80 example-ipv6-host`
- `A` — “Aggressive” scan: shorthand for OS detection, version detection, script scan and traceroute (`-O -sV -sC --traceroute`).
 - Example: `nmap -A scanme.nmap.org`

- Very noisy but convenient for single-host reconnaissance.
 - `-datadir <dirname>` — use alternate Nmap data directory (fingerprints, scripts).
 - Example: `nmap --datadir ./nmapdata scanme.nmap.org`
 - `-send-eth / --send-ip` — force using raw ethernet frames or raw IP packets.
 - Example: `nmap --send-eth -sS 192.168.1.1`
 - `-privileged` / `-unprivileged` — force Nmap to assume privileged/unprivileged environment.
 - Example: `nmap --unprivileged -sT 192.168.1.1`
 - `-v` — print Nmap version.
 - Example: `nmap -v`
 - `-h` — print help summary (the page you pasted).
 - Example: `nmap -h`
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Practical example walkthroughs (step-by-step)

1. Quick discovery of live hosts on a LAN:

- Command: `nmap -sn 192.168.1.0/24`
- What it does: sends discovery probes (ICMP/TCP/ARP depending on privileges) and lists live hosts only. No port scans performed.

2. Fast top-100 TCP ports with version detection:

- Command: `nmap --top-ports 100 -sV scanme.nmap.org`
- What it does: scans the 100 most common ports and attempts to identify services/versions.

3. Aggressive single-host scan (OS, versions, scripts, traceroute):

- Command: `nmap -A scanme.nmap.org`
- What it does: everything in `-A` —good for quick in-depth look but loud.

4. UDP + TCP full ports with timing tuned (slow):

- Command: `nmap -sS -sU -p1-65535 -T2 192.168.1.1`
- What it does: scans all TCP & UDP ports; UDP will be slow — use `T2` to reduce speed to be conservative.

5. Save outputs in multiple formats for later parsing:

- Command: `nmap -oA myscan -sV 192.168.1.0/24`
- What it does: writes `.nmap`, `.xml`, `.gnmap` versions of the scan results.

6. Run specific NSE vulnerability scripts (example: smb vuln):

- Command: `nmap --script smb-vuln* -p445 192.168.1.1`
- What it does: runs all smb-vuln scripts against SMB port 445 to check common vulnerabilities.

7. Stealth scan using SYN & decoys (only with permission):

- Command: `nmap -sS -D 1.2.3.4,5.6.7.8,ME 192.168.1.1`
- What it does: SYN scan with decoys — makes logs ambiguous. Again, use ONLY in authorized tests.

Tips & best practices

- Use `T4` for fast local scans; `T3` default for reliability across networks.
 - For UDP scans add `-max-retries` and increase timeouts: UDP is lossy and slow.
 - Combine `oA` and `-reason` to retain parsable outputs with explanations.
 - Use `-script` selectively; some `vuln` scripts can be intrusive — prefer `-script-args` and test safely.
 - Always get explicit permission before scanning networks you do not own.
 - Start with `sL` or `sn` if you only want to enumerate targets without causing heavy load.
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